

# Heart Transplant Rejection Gene Expression Profiling (AlloMap)

ACG: A-0623 (AC)

[Link to Codes](#)

**MCG Health**  
Ambulatory Care  
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## Clinical Indications for Procedure

- Heart transplant rejection gene expression profiling (AlloMap) may be indicated when **ALL** of the following are present(1)(2)(3):
  - Need for monitoring of acute cellular rejection, as indicated by **ALL** of the following:[N](#)
    - Allograft function stable, as indicated by **ALL** of the following:
      - Left ventricular ejection fraction equal to or greater than 45%
      - No evidence of heart failure
      - No history of severe allograft vasculopathy[A](17)
    - Low probability of moderate or severe acute cellular rejection[B]
    - Six months to five years following transplant
  - No prior history of antibody-mediated rejection(18)

## Alternatives to Procedure

- Alternatives include monitoring for rejection with endomyocardial biopsy.(4)(6)

## Evidence Summary

### Background

Endomyocardial biopsy is the standard method for early detection and treatment of acute rejection following heart transplant.(4)(5)(6) **(EG 2)** However, endomyocardial biopsy is an invasive procedure associated with risk, albeit low, of morbidity and mortality and subject to sampling error and interobserver variability.(7)(8)(9) **(EG 2)** Gene expression profiling encompasses different methods of quantifying relative levels of messenger RNA for large numbers of genes in specific cells or tissues. In the case of heart transplant, profiling of genes expressed by peripheral blood mononuclear cells enables quantitative assessments of intracellular messenger RNA whose levels fluctuate based on systemic immune responses.(3) **(EG 2)** Studies have shown that certain gene expression profiles correlate with histologic findings on endomyocardial biopsy, specifically, that molecular expression testing distinguishes the quiescent state from acute rejection.(2) **(EG 2)**

AlloMap, a commercially available gene expression profile for heart transplant patients, uses real-time polymerase chain reaction technology to measure the relative expression of 20 genes (11 rejection-associated genes and 9 control genes) in peripheral blood

mononuclear cells. A proprietary algorithm is then applied to the results to generate a score, ranging from 0 to 40, which is used to predict the likelihood of rejection.(10)(11) **(EG 2)**

## Criteria

The evidence for the clinical indications found in this guideline includes 15 published peer reviewed articles and 2 specialty society or other evidence-based guidelines.

For need for monitoring of acute cellular rejection, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** The clinical utility of the AlloMap test derives from its high negative predictive value (ie, ruling out acute cellular rejection).(2)(13) **(EG 2)** Low variability of sequential gene expression profile scores has been shown to increase the negative predictive value of future risk of a significant clinical event, such as graft failure or retransplant.(14) **(EG 2)** An industry-sponsored, multicenter, prospective, observational study of 1504 post-transplant patients and 2456 paired endomyocardial biopsies found AlloMap to have a negative predictive value of 98% and a positive predictive value of 2.8% in the first 6 months after the transplant; similar results were obtained at more than 6 months post transplant. Limitations included potential selection bias and gaps in data collection, such as missing biopsy dates.(15) **(EG 2)** An industry-sponsored, multicenter, prospective, observational study of 499 post-transplant patients and 938 paired endomyocardial biopsies found AlloMap to have a negative predictive value ranging from 98% to 100% and a positive predictive value ranging from 2.0% to 4.7%, which was thought to support the clinical validity of gene expression profiling; similar results were obtained both at between 2 months and 6 months and at more than 6 months post transplant.(2) **(EG 2)** An industry-sponsored multicenter randomized study of post-transplant patients demonstrated that monitoring with gene expression profiling was not associated with an increased risk of serious adverse outcomes (ie, noninferiority trial) and resulted in significantly fewer biopsies when compared with monitoring with endomyocardial biopsy; however, the authors cautioned that the results need to be interpreted in the context of significant limitations, including the selection of low-risk patients who had received a cardiac transplant more than 6 months previously, and the lack of blinding in the study, which may have influenced the intensity of immunosuppression in the gene-profiling group.(1) **(EG 1)** An industry-sponsored single-center pilot study of 60 patients who were randomized to either gene expression profiling or endomyocardial biopsy starting 55 days post transplant reported a similar primary composite endpoint (rejection with hemodynamic compromise, graft dysfunction, death, retransplant) between the 2 groups at 18 months. However, the study was not adequately powered to detect statistically significant differences between the 2 groups, and the authors recommended additional larger, randomized trials to confirm these findings.(11) **(EG 2)**

## Inconclusive or Non-Supportive Evidence

For antibody-mediated rejection, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** The AlloMap gene expression profiling test does not rule out episodes of mild rejection detected by endomyocardial biopsy and has not been validated in antibody-mediated (humoral) rejection accompanied by hemodynamic compromise.(2)(3)(12) **(EG 2)**

## Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment across patients.

## Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

**Code of Federal Regulations (CFR):** 42 CFR 422.101(19); 42 CFR 419.22(20)

**Internet-Only Manual (IOM) Citations:** CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(21); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(22); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(23)

**Medicare Coverage Determinations:** Medicare Coverage Database(24)

## References

1. Pham MX, et al. Gene-expression profiling for rejection surveillance after cardiac transplantation. New England Journal of Medicine 2010;362(20):1890-900. DOI: 10.1056/NEJMoa0912965. [ Context Link 1, 2 ] View abstract...

2. Crespo-Leiro MG, et al. Clinical usefulness of gene-expression profile to rule out acute rejection after heart transplantation: CARGO II. *European Heart Journal* 2016;37(33):2591-2601. DOI: 10.1093/eurheartj/ehv682. [ Context Link 1, 2, 3, 4, 5 ] View abstract...
3. Deng MC. The AlloMap(TM) genomic biomarker story: 10 years after. *Clinical Transplantation* 2017;31(3):e12900. DOI: 10.1111/ctr.12900. [ Context Link 1, 2, 3 ] View abstract...
4. Leone O, et al. 2011 consensus statement on endomyocardial biopsy from the Association for European Cardiovascular Pathology and the Society for Cardiovascular Pathology. *Cardiovascular Pathology* 2012 Jul-Aug;21(4):245-274. DOI: 10.1016/j.carpath.2011.10.001. (Reaffirmed 2023 Jul) [ Context Link 1, 2 ] View abstract...
5. Kiamanesh O, Toma M. The state of the heart biopsy: a clinical review. *CJC Open* 2021;3(4):524-531. DOI: 10.1016/j.cjco.2020.11.017. [ Context Link 1 ] View abstract...
6. Velleca A, et al. The International Society for Heart and Lung Transplantation (ISHLT) guidelines for the care of heart transplant recipients. *Journal of Heart and Lung Transplantation* 2023;42(5):e1-e141. DOI: 10.1016/j.healun.2022.10.015. [ Context Link 1, 2, 3 ] View abstract...
7. Crespo-Leiro MG, et al. Concordance among pathologists in the second Cardiac Allograft Rejection Gene Expression Observational Study (CARGO II). *Transplantation* 2012;94(11):1172-1177. DOI: 10.1097/TP.0b013e31826e19e2. [ Context Link 1 ] View abstract...
8. Cooper LT, et al. The role of endomyocardial biopsy in the management of cardiovascular disease: a scientific statement from the American Heart Association, the American College of Cardiology, and the European Society of Cardiology. Endorsed by the Heart Failure Society of America and the Heart Failure Association of the European Society of Cardiology. *Journal of the American College of Cardiology* 2007;50(19):1914-1931. DOI: 10.1016/j.jacc.2007.09.008. (Reaffirmed 2022 May) [ Context Link 1 ] View abstract...
9. Zhuo DX, Ginder K, Hardin EA. Markers of immune function in heart transplantation: implications for immunosuppression and screening for rejection. *Current Heart Failure Reports* 2021;18(2):33-40. DOI: 10.1007/s11897-020-00499-3. [ Context Link 1 ] View abstract...
10. HeartCare. [Internet] CareDx, Inc. Accessed at: <https://caredx.com/products-and-services/transplant-services/heart/heartcare/>. [accessed 2023 Sep 11] [ Context Link 1 ]
11. Kobashigawa J, et al. Randomized pilot trial of gene expression profiling versus heart biopsy in the first year after heart transplant: early invasive monitoring attenuation through gene expression trial. *Circulation: Heart Failure* 2015;8(3):557-564. DOI: 10.1161/CIRCHEARTFAILURE.114.001658. [ Context Link 1, 2 ] View abstract...
12. Nguyen VP, Kobashigawa JA. Antibody-mediated rejection after heart transplantation: diagnosis and clinical implications. *Current Opinion in Organ Transplantation* 2020;25(3):248-254. DOI: 10.1097/MOT.0000000000000754. [ Context Link 1 ] View abstract...
13. Austin BA, Arnold PJ, Kao A. The impact of time post cardiac transplant on gene expression profile scores, an analysis of 32,043 tests. *Journal of Cardiovascular Diseases & Diagnosis* 2013;1(3):114. [ Context Link 1 ]
14. Crespo-Leiro MG, et al. Performance of gene-expression profiling test score variability to predict future clinical events in heart transplant recipients. *BMC Cardiovascular Disorders* 2015;15:120. DOI: 10.1186/s12872-015-0106-1. [ Context Link 1 ] View abstract...
15. Moayed Y, et al. Risk evaluation using gene expression screening to monitor for acute cellular rejection in heart transplant recipients. *Journal of Heart and Lung Transplantation* 2019;38(1):51-58. DOI: 10.1016/j.healun.2018.09.004. [ Context Link 1 ] View abstract...
16. Chih S, Chong AY, Mielniczuk LM, Bhatt DL, Beanlands RS. Allograft vasculopathy: the Achilles' heel of heart transplantation. *Journal of the American College of Cardiology* 2016;68(1):80-91. DOI: 10.1016/j.jacc.2016.04.033. [ Context Link 1 ] View abstract...
17. Mavrogeni SI, Athanasopoulos G, Gouziouta A, Leontiadis E, Adamopoulos S, Kolovou G. Cardiac transplantation: towards a new noninvasive approach of cardiac allograft rejection. *Expert Review of Cardiovascular Therapy* 2017;15(4):307-313. DOI: 10.1080/14779072.2017.1307734. [ Context Link 1 ] View abstract...
18. Clerkin KJ, Restaino SW, Zorn E, Vasilescu ER, Marboe CC, Mancini DM. The effect of timing and graft dysfunction on survival and cardiac allograft vasculopathy in antibody-mediated rejection. *Journal of Heart and Lung Transplantation* 2016;35(9):1059-1066. DOI: 10.1016/j.healun.2016.04.007. [ Context Link 1 ] View abstract...
19. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2023 Jun 05 [accessed 2023 Oct 30] Accessed at: <https://www.ecfr.gov/>. [ Context Link 1 ]
20. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2023 Oct 30] Accessed at: <https://www.ecfr.gov/>. [ Context Link 1 ]
21. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 14 - Medical Devices Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2023 Oct 02] [ Context Link 1 ]
22. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12171 [Internet] Centers for Medicare and Medicaid Services. 2023 Aug 03 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2023 Oct 25] [ Context Link 1 ]
23. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 16 - General Exclusions From Coverage Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: <https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/>. [accessed 2023 Oct 02] [ Context Link 1 ]
24. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: <https://www.cms.gov/medicare-coverage-database/search.aspx?> Updated 2023 [accessed 2023 Oct 20] [ Context Link 1 ]

## Footnotes

[A] Cardiac allograft vasculopathy is characterized by coronary artery stenosis that occurs as a result of endothelial inflammation triggered by immune and nonimmune factors.(16) [ A in Context Link 1 ]

# Codes

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